

G11-03-GATE-0-NOVELTY-AUDIT-v1: Is there ancient object without paying G9 price?

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$G7 = FROZEN$, $G8 = CLOSED$, $G9 = BLOCKED$ (2 gates), $G10 = CLOSED$, $G11-01/02 = CLOSED$

$G11-03 = \text{alternative ancient rigidity}$, $GATE\ 0 = novelty$

$\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}, C_F^{conc}, C_I^{freq}, C_J^{neg}\}$, $T^* < \infty$

RULE : NO puede ser G9 con otra etiqueta

1 What G9 blocked explicitly

G9 attempted:

$$u_n(y, s) = r_n u(x_n + r_n y, t_n + r_n^2 s), \quad r_n \rightarrow 0, \quad (x_n, t_n) \rightarrow (x^*, T^*)$$

Requires: (i) uniform $L^2(B_R)$ bound $\iff r_n^{-1} E_0 \rightarrow \infty$ blocked, (ii) $u_n \rightharpoonup \bar{u}$ preserving $\|\bar{u}\|_{L^3(B_1)} \geq \varepsilon_0$ blocked: weak does not preserve lower bound.

Thus any construction using same rescaling + same compactness + same preservation is $\subset G9$.

$G11-03 \subset G9 \Rightarrow G11-03 = CLOSED/BLOCKED$

2 Three questions

A: Existence of ancient construction NOT depending on blocked compactness?

Candidate alternative procedures: - Profile decomposition without $r_n^{-1} E_0$ uniform? Needs L^3 critical profile decomposition: still requires L^2 localization? - Backward self-similar extraction via Leray scaling + Liouville? Requires Type I bound $r_n \|u\|_\infty \lesssim 1$ - new hyp, not in $\mathcal{C}_{PROVEN}$. - Global compactness via C_E^{weak} + frequency C_I^{freq} ? $N_c \rightarrow \infty$ prevents uniform bound on high frequencies, opposite direction. - Using C_J^{neg} loss of direction to get ancient? Loss gives less control, not more.

Check: To get $\mathcal{A} \neq \emptyset$ ancient, need space-time window $(-\infty, 0]$ limit. All known procedures for NS ancient from $T^* < \infty$ use blow-up scaling $r_n \rightarrow 0$ plus uniform $L^2(B_R)$ bound. Without it, $E_0 r_n^{-1}$ diverges. No known alternative global structure (e.g., monotonicity) gives ancient without scaling, as G11-01 already BLOCKED.

Thus: $\exists$ construction without paying G9 price?

B: Non-trivial property surviving?

Even if $\mathcal{A} \neq \emptyset$ via other route, need non-trivial property not requiring direct C_F^{conc} preservation. E.g., $\|\bar{u}\|_{L^2(B_1)} \geq c > 0$, or enstrophy non-zero, or $\bar{u} \neq 0$. If construction yields only $\bar{u} = 0$ possible, no contradiction: $\bar{u} = 0$ is compatible with $T^* < \infty$ (blow-up profile could vanish weakly). Need property that survives despite weak convergence, but not same $\|L^3 \geq \varepsilon_0$ which already killed in G9.

C: Rigidity theorem applicable to exact class?

If A+B YES, need classification $\mathcal{A} = \{0\}$ or Liouville. Known rigidities: ancient mild bounded $\in L_t^\infty L_x^3$ trivial? Requires bounded L^3 - not from $\mathcal{C}_{PROVEN}$? ESS rigidity for ancient self-similar? Needs self-similar hypothesis. No.

3 Gate 0 audit result

A: No construction known from $\mathcal{C}_{PROVEN}$ that produces ancient object without using critical compactness blocked in G9. All $T^* < \infty \Rightarrow$ ancient routes for NS in literature use r_n scaling + uniform $L^2(B_R)$ bound. That bound is exactly $r_n^{-1}E_0$ which we proved $\rightarrow \infty$. Therefore A fails with current C_i .

Since G11-01 monotone quantity already BLOCKED, cannot produce ancient via monotone global route. G11-02 topology BLOCKED, cannot produce ancient via topological persistence.

Hence:

$$A = NO \text{ with } \mathcal{C}_{PROVEN}$$

$$G11-03-GATE-0 = BLOCKED$$

$$G11-03 = CLOSED/BLOCKED \text{ as subset of } G9$$

This is structural closure: G9 did not just block one attempt, it blocks whole class of blow-up compactness constructions because the obstruction $r_n^{-1}E_0 \rightarrow \infty$ + weak losing ε_0 is intrinsic to L^2 -based localization, not to particular subsequence.

If future new bound beyond $\mathcal{C}_{PROVEN}$ appears (e.g., new scale-critical uniform bound from $T^* < \infty$ alone), Gate 0 could be revisited. With current $\mathcal{C}_{PROVEN}$:

$$G7 \rightarrow G8 \rightarrow G9 \rightarrow G10 \rightarrow G11-01 \rightarrow G11-02 \rightarrow G11-03 = FROZEN \rightarrow CLOSED \rightarrow BLOCKED \rightarrow CLOSE$$

$$Frontier \text{ map with current } C_i = EXHAUSTED$$

$$STOP = PROGRESO : no candidate family remains unaudited$$